

Tesla Stock Price Prediction for January 2024

Comparing linear regression models with stock
and NLP features

Build Fellowship: Predicting Stock Price in a Quantitative Way

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Table of Contents

Introduction

Data Collection

Feature Engineering

Model Training

Evaluation

Prediction

Insights

Introduction

- Goal: predict Tesla closing prices for January 2024 using 2023 historical data
- Compare linear regression models with stock-only and news/NLP-enhanced features

Data Collection

- Tesla stock data source: **yfinance**
- Tesla-related news data: **NBC News** and **The New York Times**

Final cleaned datasets used for modeling

Source	Period	Key Columns	Rows
Tesla stock	2023-01-01 to 2024-01-31	Date, Close, High, Low, Open, Volume	271
NBC + NYT news	2023-01-01 to 2024-01-31	date, title, summary	249

Stock data loading:

```

ticker = "TSLA"
data = yf.download(ticker, start="2023-01-01", end="2024-02-01")
data.reset_index(inplace=True)

```

News data loading:

```

news = pd.read_csv("combined_tesla_news_clean.csv")
news["date"] = pd.to_datetime(news["date"], errors="coerce").dt.floor("D")

```

Feature Engineering

STOCK FEATURES:

- Open-close price difference
- Daily high-low range
- Trading volume change
- Squared high-low range
- Lagged closing-price features
- Simple and exponential moving averages

Stock feature creation:

```
data["Open_Close_Diff"] = data["Open"] - data["Close"]
data["High_Low_Range"] = data["High"] - data["Low"]
data["Volume_Change_Pct"] = data["Volume"].pct_change()
data["Squared_High_Low_Range"] = data["High_Low_Range"] ** 2

data["Close_lag_1"] = data["Close"].shift(1)
data["Close_lag_5"] = data["Close"].shift(5)

data["SMA_10"] = data["Close"].rolling(window=10).mean()
data["SMA_20"] = data["Close"].rolling(window=20).mean()

data["EMA_10"] = data["Close"].ewm(span=10, adjust=False).mean()
data["EMA_20"] = data["Close"].ewm(span=20, adjust=False).mean()
```

NEWS/NLP FEATURES:

- Daily article count
- Daily average sentiment
- Lagged article count / lagged sentiment
- TF-IDF text features

Sentiment scoring:

```
news["vader_sentiment"] = news["text_for_nlp"].apply(
    lambda x: sia.polarity_scores(x)["compound"]
)
```

Daily aggregation and lagged features:

```
daily_sentiment = news.groupby("date").agg(
    article_count=("text_for_nlp", "count"),
    vader_avg=("vader_sentiment", "mean"),
    vader_max=("vader_sentiment", "max"),
    vader_min=("vader_sentiment", "min")
).reset_index()

daily_sentiment["article_count_lag_1"] = daily_sentiment["article_count"].shift(1)
daily_sentiment["textblob_avg_lag_1"] = daily_sentiment["textblob_avg"].shift(1)
daily_sentiment["vader_avg_lag_1"] = daily_sentiment["vader_avg"].shift(1)
```

TF-IDF vectorization:

```
vectorizer = TfidfVectorizer(
    lowercase=True, stop_words="english", max_features=300, ngram_range=(1, 2)
)
```

Model Training

ESTIMATOR:

- Linear Regression

TARGET VARIABLE:

- Tesla daily closing price (Close)

TRAINING AND EVALUATION:

- 2023 data

FINAL PREDICTION PERIOD:

- January 2024

FEATURE SETS TESTED:

- **Stock-only models**
 - Base
 - SMA-based lag
 - EMA-based lag
- **News/NLP-enhanced models**
 - TextBlob
 - VADER
 - TF-IDF

Training workflow:

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42  
)  
  
model = LinearRegression()  
model.fit(X_train, y_train)  
  
y_pred = model.predict(X_test)
```

Evaluation

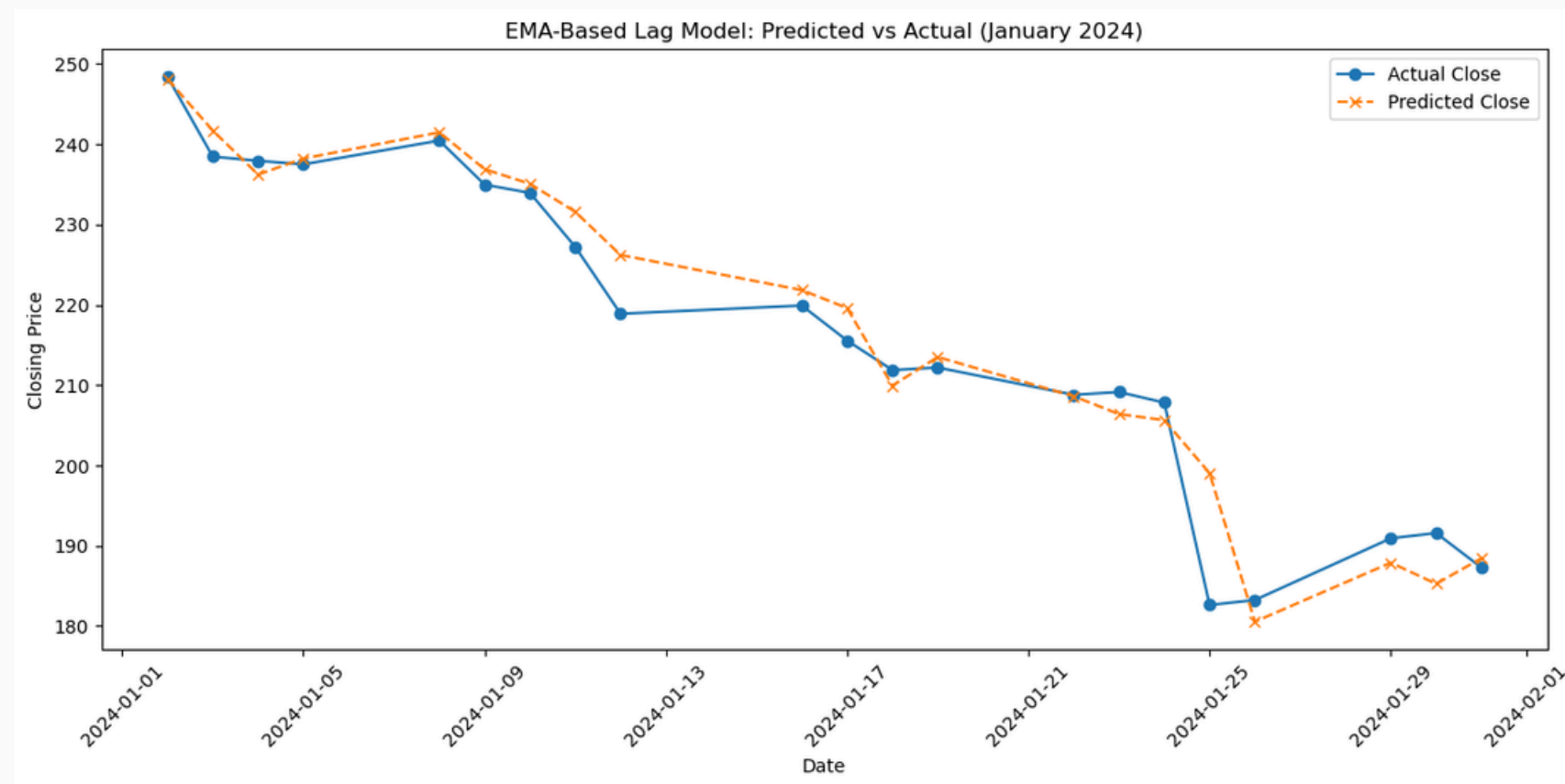
- On the 2023 test split, the EMA-based lag model performed best overall.
- For January 2024 prediction, some NLP-based models performed slightly better.

Model	Test_R ²	Test_MSE	Jan_R ²	Jan_MSE
Base	0.0912	1432.4078	-0.5639	622.4621
Lag (SMA)	0.9877	13.0181	0.9346	26.0316
Lag (EMA)	0.9921	11.9193	0.9452	21.7958
Linear Regression + TF-IDF (SMA)	0.9863	14.4168	0.9377	24.7954
Linear Regression + TF-IDF (EMA)	0.9897	15.4216	0.9503	19.7967
Linear Regression + TextBlob (SMA)	0.9852	15.5903	0.9301	27.8278
Linear Regression + TextBlob (EMA)	0.9913	13.0077	0.9451	21.8617
Linear Regression + VADER (SMA)	0.987	13.7566	0.9343	26.1486
Linear Regression + VADER (EMA)	0.9914	12.8658	0.9463	21.367

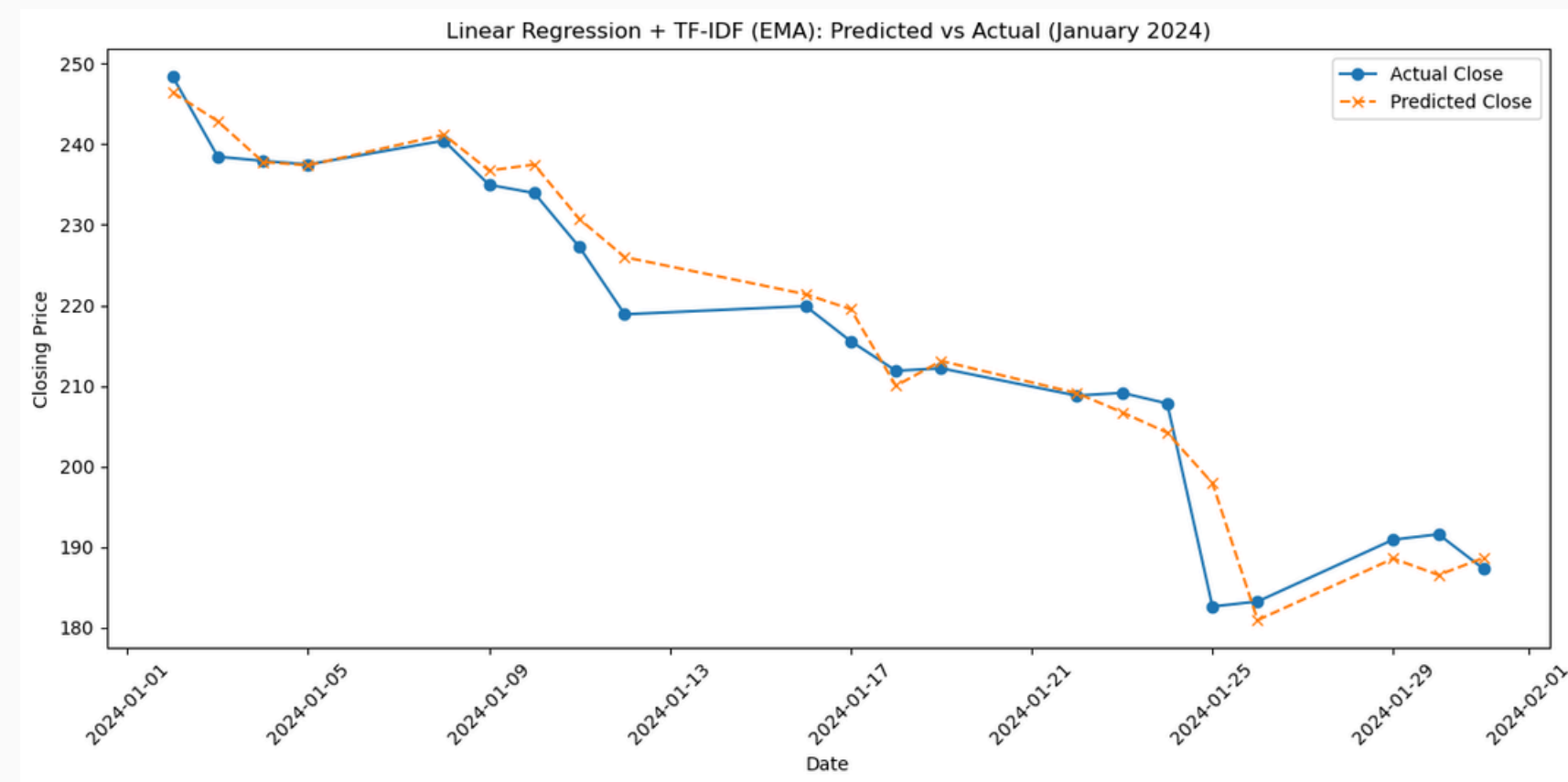
Prediction

- Best on 2023 test split: EMA-Based Lag Model
- Best on January 2024 prediction: Linear Regression + TF-IDF (EMA)

LINEAR REGRESSION WITH EMA-BASED LAG FEATURES



LINEAR REGRESSION WITH TF_IDF NEWS FEATURES (EMA)



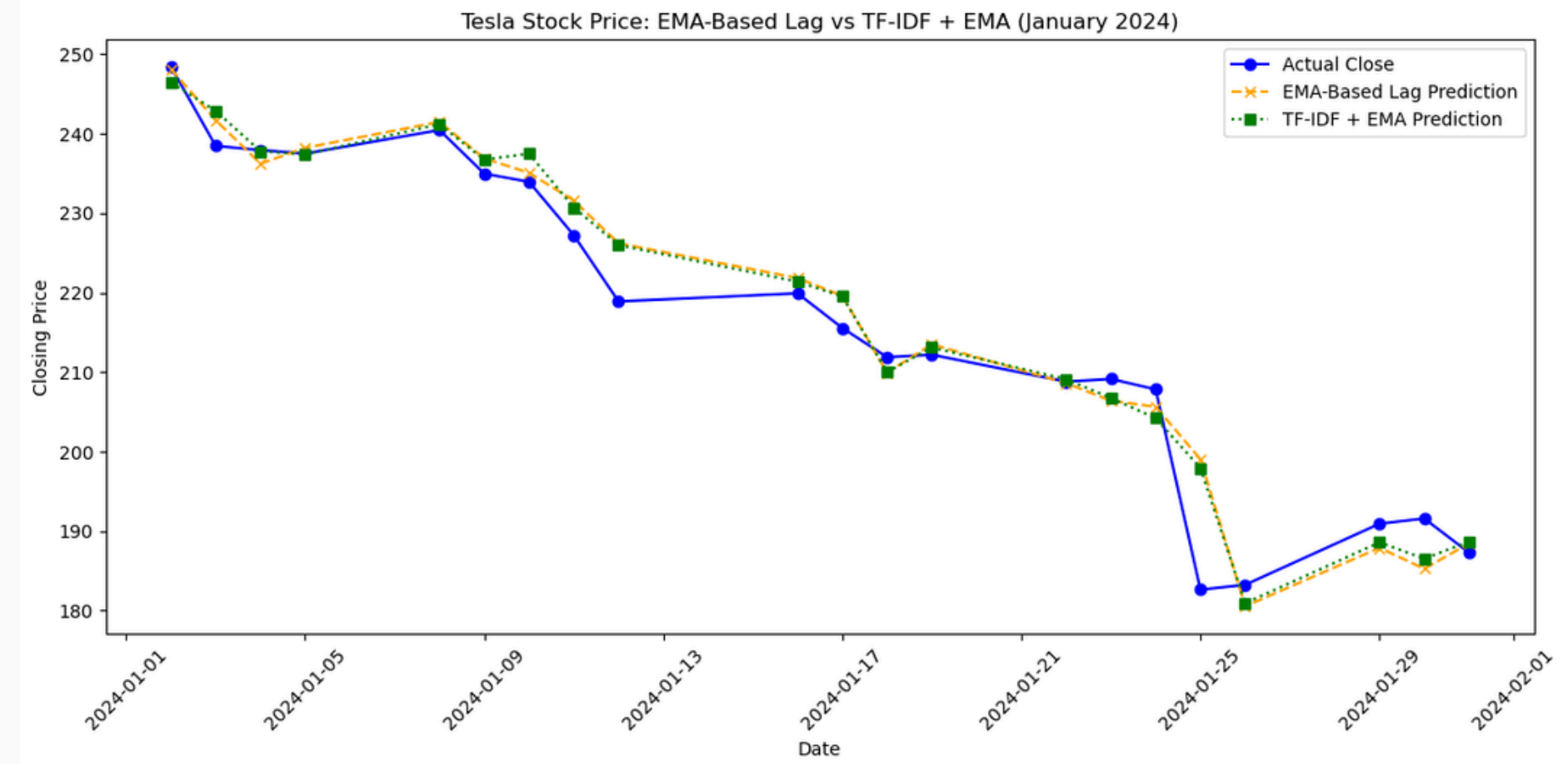
Insights

INSIGHTS:

- **EMA** performed better than SMA.
- **TF-IDF (EMA)** gave the best January 2024 prediction.
- NLP features helped improve short-term prediction.
- Stock-history features were still strong predictors.

FUTURE WORK:

- Use more finance-focused news sources
- Improve NLP feature engineering
- Align news timing with market hours





Thank you

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